

Intelligent SQL Agent — Documentation

Delivery: v1.1 · **Repository:** <https://github.com/pradipkundu144/intelligent-sql-agent>

A natural-language interface to a business analytics database. Analysts type a question in plain English; the agent understands intent, generates read-only SQL, validates it against three independent safety layers, executes against Postgres, and returns a plain-English answer with the SQL visible for verification.

Quick links

TOPIC	DOCUMENT	AUDIENCE
Start here — what was built and why	01_ARCHITECTURE	Everyone
Run the app locally	02_SETUP	Reviewer, DevOps
Positive / negative / neutral / multi-intent examples	03_SCENARIOS	Reviewer
Metrics, RAG A/B result, performance improvements	04_EVALUATION	Reviewer, Engineering lead
LangGraph vs LangChain vs Google ADK	05_FRAMEWORK_COMPARISON	Engineering lead
Finish-line, feature, and scale roadmap	06_EXTENSION_ROADMAP	Product, Engineering lead
Commercial models and moat	07_MONETISATION	Business stakeholders
Key design decisions	08_DESIGN_DECISIONS	Engineering lead, DevOps

Headline metrics

All headline numbers from the automated evaluation harness. Reproducible via `docker compose exec backend python -m eval.run` or the "Run Evaluation" button in the admin dashboard.

METRIC	TARGET	ACTUAL
Execution accuracy (positive set, 15 cases)	≥ 90%	100% (15/15)
Guardrail catch rate (negative set, 10 cases)	100%	100% (10/10)
Graceful handling (neutral set, 6 cases)	≥ 80%	100% (6/6)
RAGAS faithfulness — mean	≥ 0.85	0.87
RAGAS answer relevancy — mean	≥ 0.85	0.88
RAG accuracy lift (measured A/B)	—	\$86,670 delta on the "revenue" question

Full breakdown in 04_EVALUATION .

What makes this different from a demo

- **Destructive queries are impossible by construction** — three independent safety layers, the load-bearing one being a Postgres role with every write privilege revoked. See 01_ARCHITECTURE §2.5.
- **Every request is fully observable** — per-stage latency, token counts, cost, plus deep-link to Langfuse and LangSmith traces. In-app TracePanel on every answer.
- **Quality is measured, not asserted** — four automated metrics across 31 test cases. See 04_EVALUATION .
- **RAG earns its keep** — one curated context row about business conventions swung total-revenue accuracy by \$86,670 on the demo dataset. Measured directly. See 04_EVALUATION §3.

Reading order

- **30 minutes:** README → 01_ARCHITECTURE → 04_EVALUATION
- **60 minutes:** add 03_SCENARIOS + 08_DESIGN_DECISIONS
- **Full read:** all nine in order

For the DevOps team

Two supporting documents (**DOCKER_COMPOSE_STACK** and **PRODUCTION_AND_SCALING**) live in the source repository's `infra/` folder and can be provided separately on request. They cover the container stack walkthrough and the day-one production must-changes + scaling axes + readiness tier checklists.

Safety invariants that must be preserved through deployment are called out in 08_DESIGN_DECISIONS .